CO2 AT1

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Course Code/Name: CSA1613- Data Warehousing and Data Mining

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1. Missing Value Imputation using Mean Method (Hospital Patient Records)

Given Data

95, 102, 110, –, 120, 115, 108, –, 130, 125

Step 1: Calculate the Mean of Available Values

Available values:

95, 102, 110, 120, 115, 108, 130, 125

Sum = 95 + 102 + 110 + 120 + 115 + 108 + 130 + 125 = 905

Number of available observations = 8

$$\text{Mean} = \frac{905}{8} = 113.125$$

Mean = 113.125 mg/dL

Step 2: Replace Missing Values

Both missing values are replaced with 113.125.

Step 3: Completed Dataset

95, 102, 110, 113.125, 120, 115, 108, 113.125, 130, 125

2. Min–Max Normalization of Cholesterol Levels

Given Data

160, 180, 200, 220, 240, 260

Minimum = 160 Maximum

= 260

Formula:

$$x' = \frac{x - \min}{\max - \min}$$

$$x' = \frac{x - 160}{260 - 160} = \frac{x - 160}{100}$$

Cholesterol (mg/dL)	Calculation	Normalized Value
160	(160−160)/100	0.00
180	(180−160)/100	0.20
200	(200−160)/100	0.40
220	(220−160)/100	0.60
240	(240−160)/100	0.80
260	(260−160)/100	1.00

Final Normalized Dataset

0.00, 0.20, 0.40, 0.60, 0.80, 1.00

3. Decimal Scaling Normalization of Annual Medical Expenses

Given Data

8500, 12000, 15000, 18000, 22500, 30500, 47000

Step 1: Determine the value of j Maximum value = 47000

Decimal scaling formula: $x' = \frac{x}{10^j}$

Choose the smallest j such that

$$\frac{47000}{10^j} < 1$$

Since

47000 has 5 digits, $j = 5$

Step 2: Normalize Each Observation

Original Value (₹)	Calculation	Normalized Value
8500	8500/100000	0.085
12000	12000/100000	0.120
15000	15000/100000	0.150
18000	18000/100000	0.180
22500	22500/100000	0.225
30500	30500/100000	0.305
47000	47000/100000	0.470

Final Normalized Dataset

0.085, 0.120, 0.150, 0.180, 0.225, 0.305, 0.470

Step 3: Why Decimal Scaling is Useful •

Reduces the magnitude of large numbers.

- Keeps all values within a small range.
- Preserves the relative ordering of data.
- Makes data suitable for machine learning and data mining algorithms.
- Improves computational efficiency.

4. Correlation Analysis between Exercise Hours and BMI

Given Data

Exercise Hours (x)	BMI (y)
2	31

4	29
5	28
6	26
8	24

Step 1: Calculate Means

Mean of x

$$\bar{x} = \frac{2 + 4 + 5 + 6 + 8}{5} \quad \frac{25}{5} == 5$$

Mean of y

$$\bar{y} = \frac{31 + 29 + 28 + 26 + 24}{5} \quad \frac{138}{5} == 27.6$$

Step 2: Pearson Correlation Coefficient

x	**y**	**$x - \bar{x}$**	**$y - \bar{y}$**	**$(x - \bar{x})(y - \bar{y})$**	**$(x - \bar{x})^2$**	**$(y - \bar{y})^2$**
2	31	-3	3.4	-10.2	9	11.56
4	29	-1	1.4	-1.4	1	1.96
5	28	0	0.4	0	0	0.16
6	26	1	-1.6	-1.6	1	2.56
8	24	3	-3.6	-10.8	9	12.96

Totals:

$$\sum(x - \bar{x})(y - \bar{y}) = -24$$

$$\sum(x - \bar{x})^2 = 20$$

$$\sum(y-\bar{y})^2 = 29.2$$

Pearson formula:

$$r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \sum(y-\bar{y})^2}}$$

$$r = \frac{-24}{\sqrt{20 \times 29.2}}$$

$$r = \frac{-24}{24.17}$$

$$\boxed{r \approx -0.99}$$

Step 3: Interpretation

- Pearson correlation coefficient = −0.99
- There is a very strong negative correlation between exercise hours and BMI.
- As exercise hours increase, BMI decreases.

5. Equal-Depth Binning using Bin Means

Given Sorted Data

10, 12, 15, 18, 20, 22, 25, 28, 30, 35, 40, 45

Total observations = 12

Number of bins = 4

Each bin contains

$$12/4 = 3 \text{ values}$$

Bin 1

10, 12, 15

Mean = $\frac{10+12+15}{3} = \frac{37}{3} = 12.33$

Smoothed Bin

12.33, 12.33, 12.33

Bin 2

18, 20, 22

Mean = $\frac{18+20+22}{3} = 20$

Smoothed Bin

20, 20, 20

Bin 3

25, 28, 30

Mean = $\frac{25+28+30}{3} = \frac{83}{3} = 27.67$

Smoothed Bin

27.67, 27.67, 27.67

Bin 4

35, 40, 45

Mean = $\frac{35+40+45}{3} = 40$

Smoothed Bin

40, 40, 40

Final Smoothed Dataset

12.33, 12.33, 12.33, 20, 20, 20, 27.67, 27.67, 27.67, 40, 40, 40